Singleton intrinsic-distance tests do not characterize compact eikonal spaces

Candidate counterexample packet for arXiv:2605.20486, Question 1

11 August 2026

Status: candidate counterexample, likely valid, subject to human review. The construction below gives a full negative answer to Question 1 of Salas, Tapia-García, and Venegas M.

1 Source question

The source is David Salas, Sebastián Tapia-García, and Francisco Venegas M., *The reach and limits of slope eikonal equations in compact spaces*, arXiv:2605.20486v1 [math.FA], submitted 19 May 2026 [1]. For a metric space (X, d), its intrinsic distance is denoted by d_I, and the descent slope of a real-valued function f is

$$\mathsf{s}[f](x) = \limsup_{y \to x} \frac{[f(x) - f(y)]_+}{d(x, y)}.$$

Question 1 on printed page 25 asks:

> Let (X, d) be a compact metric space such that $d_I(\cdot, x)$ has slope 1 on $X \setminus \{x\}$ for every $x \in X$. Is (X, d) an eikonal space?

The source proves that a compact metric space is eikonal if and only if $\mathsf{s}[d_I(\cdot, K)] = 1$ on $X \setminus K$ for every nonempty closed K. It also notes that it is enough to test closed countable K. Thus a negative answer follows from a compact space on which every singleton test passes but one closed countable-set test fails.

2 Proof intuition

The obstruction is a failure of uniformity in the endpoint. Attach countably many polygonal arcs to a common root and let them accumulate on one straight limit arc. Every finite arc makes a tiny dogleg near the root and then runs almost parallel to the limit arc. For any *fixed* target point, the intrinsic-distance function only sees ordinary straight local geometry, so its descent slope is one. The dogleg is at a fixed positive scale for that fixed branch and disappears when the local-slope limit is taken.

Now put all branch endpoints into one closed set. At the root, the closest endpoint is the endpoint of the limit arc. At the post-dogleg point of branch n, the closest endpoint switches to the endpoint of branch n. This switching makes the closed-set distance fall twice as fast as the ambient Euclidean distance. The nearest endpoint varies with n, which is precisely what no individual singleton test detects.

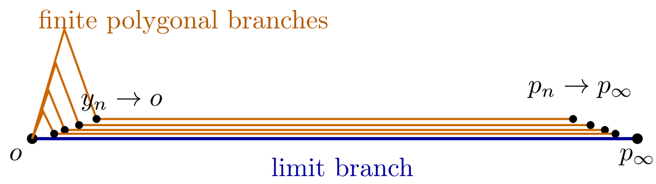

Figure 1: Schematic planar projection. In the actual construction, different finite branches lie in different half-planes in $\mathbb{R}^3$ and intersect only at the root.

3 Formal counterexample

Write $\mathbb{R}^3 = \mathbb{R} \times \mathbb{R}^2$. Let $o = (0,0)$ and $p_\infty = (1,0)$. Choose pairwise non-collinear unit vectors $v_n \in \mathbb{R}^2$, and set

$$\varepsilon_n = 2^{-n-3}, \qquad h_n = \varepsilon_n^2,$$

$$a_n = \left(\frac{\varepsilon_n}{2}, \varepsilon_n v_n\right), y_n = (\varepsilon_n, h_n v_n), p_n = (1-\varepsilon_n, h_n v_n).$$

Define the straight limit arc and the finite polygonal arcs by

$$\Gamma_\infty = [o, p_\infty], \qquad \Gamma_n = [o, a_n] \cup [a_n, y_n] \cup [y_n, p_n],$$

and let

$$X = \Gamma_\infty \cup \bigcup_{n \geq 1} \Gamma_n$$

with the metric d induced by the Euclidean norm of $\mathbb{R}^3$.

The vectors may be chosen, for example, as $v_n = (\cos(1/(n+1)), \sin(1/(n+1)))$. Distinct finite arcs then lie in distinct half-planes through the first coordinate axis, and $\Gamma_n \cap \Gamma_m = \{o\}$ for $n \neq m$, including $m = \infty$.

Let

$$\alpha_n = \operatorname{len}([o, a_n] \cup [a_n, y_n]).$$

A direct computation gives

$$\alpha_n = \varepsilon_n \left(\frac{\sqrt{5}}{2} + \sqrt{\frac{1}{4} + (1-\varepsilon_n)^2}\right) > 2\varepsilon_n. \tag{1}$$

Indeed, $\varepsilon_n \leq 1/16$, and the parenthesis is at least $\sqrt{5}/2 + 17/16 > 2$. Hence the total length of Γ_n is

$$L_n = \alpha_n + (1 - 2\varepsilon_n) > 1. \tag{2}$$

Lemma 1. *X is compact. Its intrinsic metric is the arclength tree metric on the arcs: each finite branch can be left only through o, and the intrinsic distance between two points is the length of the unique branch path joining them.*

Proof. Take any sequence in X. A subsequence lying in a fixed compact polygonal arc converges. Otherwise its branch indices tend to infinity. Points on the two dogleg segments then converge to o, because their Euclidean norms are $O(\varepsilon_n)$. Points on the horizontal tails have a subsequence whose first coordinates converge to some $t \in [0,1]$, while their transverse coordinates have norm $h_n \to 0$; they converge to $(t,0) \in \Gamma_\infty$. Thus X is sequentially compact, hence compact.

For fixed n, every point of $\Gamma_n \setminus \{o\}$ has positive distance from the other branches. Therefore $\Gamma_n \setminus \{o\}$ is relatively open in $X \setminus \{o\}$, and a continuous path leaving Γ_n must pass through o. Since different branches meet only at o, their intrinsic metric is exactly the asserted arclength tree metric. $\square$

For a point z on a finite branch, write $\sigma(z) = d_I(o, z)$ for its depth. On the tail $[y_n, p_n]$, if $z = (t, h_n v_n)$, then

$$\sigma(z) = \alpha_n + (t - \varepsilon_n) = t + c_n, \qquad c_n := \alpha_n - \varepsilon_n > 0. \tag{3}$$

Theorem 1. *For every $p \in X$,*

$$\mathsf{s}[d_I(\cdot, p)](q) = 1 \qquad (q \in X \setminus \{p\}).$$

Nevertheless, X is not an eikonal space. More precisely, the closed countable set

$$K = \{p_\infty\} \cup \{p_n : n \geq 1\}$$

satisfies

$$\mathsf{s}[d_I(\cdot, K)](o) \geq 2.$$

Consequently Question 1 of arXiv:2605.20486 has a negative answer.

Proof. Fix $p \in X$ and put $F_p = d_I(\cdot, p)$.

Finite branches. Let $q \in \Gamma_n \setminus \{o, p\}$. A sufficiently small neighborhood of q lies in Γ_n. In the arclength coordinate σ, the function F_p is either $C + \sigma$ or $|\sigma - \sigma(p)|$. On each straight segment incident to q, Euclidean distance from q equals arclength. Thus every descent quotient is at most one, and the incident segment pointing toward p along the tree realizes quotient one. This remains true at the two polygonal vertices and at the endpoint of the branch.

The root. If p lies on a finite branch, only points moving into that fixed branch can decrease F_p near o; its first segment is straight, so the quotient is one. All other branches increase the distance to p. If p lies on the limit branch, its straight segment gives the same conclusion. Hence $\mathsf{s}[F_p](o) = 1$ whenever $p \neq o$.

The limit branch. Let $q = (t, 0)$ with $t > 0$. Besides points of Γ_∞, a sequence approaching q must eventually consist of tail points $z_n = (r_n, h_n v_n)$ with $r_n \to t$ and branch indices tending to infinity. By (3), $\sigma(z_n) = r_n + c_n$ with $c_n > 0$.

If p lies on a fixed finite branch at depth b, then eventually

$$[F_p(q) - F_p(z_n)]_+ = [t - r_n - c_n]_+ \leq [t - r_n]_+ \leq d(q, z_n).$$

If $p = (b, 0)$ lies on the limit branch and $t \leq b$, then $F_p(z_n) > F_p(q)$, so there is no descent. If $t > b$, then

$$[F_p(q) - F_p(z_n)]_+ = [t - r_n - c_n - 2b]_+ \leq [t - r_n]_+ \leq d(q, z_n).$$

Thus cross-branch sequences never give slope larger than one. Moving along the straight limit branch toward p (or toward o when p is on a finite branch) realizes quotient one. This completes the singleton-slope proof.

It remains to show failure for a closed set. Compactness and $p_n \to p_\infty$ show that K is closed. Equations (2) and the unit length of Γ_∞ give

$$d_I(o, K) = 1.$$

At y_n, the closest point of K is p_n, reached along the straight tail:

$$d_I(y_n, K) = d_I(y_n, p_n) = 1 - 2\varepsilon_n.$$

Indeed, every route from y_n to any other point of K first passes through o and has length greater than one. Finally,

$$\frac{d_I(o, K) - d_I(y_n, K)}{d(o, y_n)} = \frac{2\varepsilon_n}{\sqrt{\varepsilon_n^2 + \varepsilon_n^4}} = \frac{2}{\sqrt{1 + \varepsilon_n^2}} \longrightarrow 2.$$

Therefore $\mathsf{s}[d_I(\cdot, K)](o) \geq 2$, whereas the eikonal-space characterization in the source requires this slope to equal one. Hence X is not eikonal, despite passing every singleton test. □

4 Verification and limitations

The proof has no computational dependency. The bundled checker recomputes (1), verifies $\alpha_n > 2\varepsilon_n$ and $L_n > 1$ over a large finite range, and confirms convergence of the closed-set descent quotient to two. The formal proof, rather than the finite check, establishes the result for every n.

The conclusion concerns exactly the compact singleton-test question. It does not classify noncompact eikonal spaces or other Hamilton-Jacobi equations. The counterexample is a compact subset of Euclidean three-space with its induced metric; its intrinsic metric is non-geodesically compatible with the ambient chord metric at a sequence of shrinking doglegs.

5 Novelty check and review recommendation

On 11 August 2026, the run indexes were searched for arXiv:2605.20486 and for the phrases "slope eikonal", "singleton intrinsic distance", and close variants. Bounded primary arXiv searches for the exact question, title, and singleton formulation found only source version v1 and no later answer. The source arXiv record still lists only v1. Novelty confidence is therefore moderate to high, with the usual caveat that a two-month-old problem may have unindexed private or forthcoming work.

Human review should focus on two points: (i) the assertion that a finite branch can be left only through the root, which identifies the intrinsic tree metric; and (ii) the cross-branch slope estimate on the limit arc. Both are spelled out explicitly above and are independent of computation.

A Source-page evidence

Figure 2 reproduces printed page 25 of the source PDF, including the exact open question.

References

[1] D. Salas, S. Tapia-García, and F. Venegas M., *The reach and limits of slope eikonal equations in compact spaces*, arXiv:2605.20486v1 [math.FA], 2026.

If the metric space is not compact, the classification gets more complicated. Length spaces remain eikonal, but eikonal spaces may not even be connected, and are therefore not included in neither the class of spaces where d_I is continuous nor the class of rectifiably connected spaces.

Example 6.1. Consider two parallel copies of $\mathbb{R}$ in $\mathbb{R}^2$, let us say $L_1 = \mathbb{R} \times \{0\}$ and $L_2 = \mathbb{R} \times \{1\}$, and set $X = L_1 \cup L_2$, endowed with the induced distance from $\mathbb{R}^2$. This space is clearly not rectifiably connected. However, every eikonal equation of the form (3) admits solutions. Indeed, take $\Omega \subset X$ a proper open set, a continuous function $\ell : \Omega \to (0, +\infty)$ and a continuous function $g : \partial\Omega \to \mathbb{R}$ verifying (CC). Note that in this space $\partial\Omega$ could be empty. In this case, we assume that (CC) holds by vacuity. We distinguish two cases:

1. $L_1 \not\subset \Omega$ and $L_2 \not\subset \Omega$: In this case, we just solve two independent equations with $\Omega_1 = L_1 \cap \Omega$ and $\Omega_2 = L_2 \cap \Omega$.

2. $L_1 \subset \Omega$: In this case, since Ω must be a proper subset of X, we solve the equation in $\Omega_2 = L_2 \cap \Omega$, obtaining the solution $u_2 : \overline{\Omega}_2 \to \mathbb{R}$, and define

$$u : x = (x_1, x_2) \mapsto u(x) = \begin{cases} \int_0^{x_1} \ell(t, 0)dt + c & \text{if } x \in L_1, \\ u_2(x) & \text{if } x \in \overline{\Omega_2}, \end{cases}$$

   where $c \in \mathbb{R}$ is a constant. Note that in this case there are infinitely many solutions. If $\partial\Omega = \emptyset$, that is $\Omega = L_1$, we simply set $u(x) = \int_0^{x_1} \ell(t, 0)dt + c$. The case when $L_2 \subset \Omega$ is analogous.

In any case, the eikonal equation admits at least one solution, and so this is an eikonal space. $\Diamond$

Open questions: In this manuscript we have studied in depth the slope eikonal equations in compact spaces. From this, we would like to point out the following two interesting lines of research: the study and classification of noncompact eikonal spaces, as well as, the study of well-posedness of more complicated Hamilton-Jacobi equations. On the other hand, one can observe from our proof in Section 5 that Corollary 1.4 can be improved by reducing the family of closed sets only to countable ones: a compact metric space (X, d) is eikonal if and only if $d_I(\cdot, K)$ has slope 1 on $X \setminus K$, for every $K \subset X$ closed and countable. We conjecture that this can be further reduced to singletons.

Question 1: Let (X, d) be a compact metric space such that $d_I(\cdot, x)$ has slope 1 on $X \setminus \{x\}$ for every $x \in X$. Is (X, d) an eikonal space?

References

[1] L. Ambrosio and J. Feng. On a class of first order Hamilton-Jacobi equations in metric spaces. *J. Differ. Equations*, 256(7):2194–2245, 2014.

[2] Z. M. Balogh, A. Engulatov, L. Hunziker, and O. E. Maasalo. Functional inequalities and Hamilton–Jacobi equations in geodesic spaces. *Potential analysis*, 36(2):317–337, 2012.

[3] G. Barles. An introduction to the theory of viscosity solutions for first-order Hamilton-Jacobi equations and applications. In *Hamilton-Jacobi equations: approximations, numerical*

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Figure 2: Source paper, printed page 25: the singleton-test conjecture and Question 1.